Digital Technology Access and STEM Learning in Spanish Secondary Education

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Abstract. **This paper examines how access to digital technology relates to STEM learning and education among secondary-school students in Spain. It combines official household ICT statistics from the Spanish National Statistics Institute (INE), findings from ICILS 2023, and four peer-reviewed studies using Spanish student and school data. In 2025, 97.4% of households with at least one member aged 16–74 had broadband access and 83.8% had some type of computer; these household figures do not guarantee that each student has reliable, individual access for schoolwork. In ICILS 2023, Spain scored 495 points in computer and information literacy, above the international participant average of 476 and the EU average of 493. However, 44.4% of Spanish students did not reach Level 2, and the socioeconomic performance gap was 69 points (below the EU gap of 79 points reported by INEE). Spanish studies also show that relationships between ICT use and outcomes vary by subject, with a positive association reported for science but not statistically significant effects for mathematics or reading in one PISA-based analysis. The evidence suggests that policy should combine equitable access with digital-skills development, meaningful educational use and attention to socioeconomic and regional differences.**

Keywords. digital divide; digital access; digital competence; STEM education; Spain

1. Introduction

Digital tools are used in many parts of school life, from finding information and submitting assignments to practising mathematics and exploring scientific ideas. They can give students access to simulations, online resources and software that would be difficult to use without a computer or a reliable connection. However, having a device does not automatically mean that a student can use it effectively or benefit from it in the same way as other students.

The digital divide is not only about whether a household has internet access. It can also involve the type and reliability of the connection, whether a student has a suitable device available when needed, and whether they have the skills to use technology for schoolwork. These differences matter in STEM subjects, where students may need to work with digital information, specialised applications or computer-based tasks.

Spain is a useful case because education is managed through both national and autonomous-community institutions. The OECD's review of Spain's digital education ecosystem explains that responsibility for much school digital infrastructure is devolved to the autonomous communities. Regional authorities can therefore differ in the tools and resources they provide, even though the national government also supports digital education and provides resources across regions (OECD, 2023). This governance structure is relevant to how digital learning resources are distributed.

This paper asks: to what extent does access to technology relate to students' participation in STEM learning and education in Spanish secondary schools? It argues that access is an important condition for digital learning, but its relationship with educational outcomes depends on digital competence, socioeconomic circumstances, how technology is used and the school context. It distinguishes evidence of association from evidence that can support a causal claim.

2. Literature Review

Fernández-Gutiérrez, Giménez and Calero (2020) examined school ICT use and student outcomes in mathematics, reading and science using PISA data from 2009, 2012 and 2015. Their analysis included 61,042 students from 2,195 schools across 13 Spanish autonomous communities. They found that greater school ICT use had a positive effect on science outcomes, while the estimated effects on mathematics and reading were not statistically significant. This challenges the assumption that increasing technology use improves all subjects equally and makes the regional context central to the analysis.

Ortega-Rodríguez (2025) studied predictors of computational-thinking performance using PISA 2022 data from 30,800 Spanish secondary students in 966 schools. The study used multilevel regression, analysing factors at both student and school level. Its results identified socioeconomic context and ICT use at home as important predictors. At school level, private schools and schools with more ICT resources scored higher than public schools with fewer resources. These findings connect technology and computational

thinking, a skill relevant to STEM, but they do not show that resources alone caused the differences. School type and socioeconomic background may be related to other advantages that also affect outcomes.

Niño-Cortés, Grimalt-Álvaro, Serrano and Sanabria (2024) examined digital-divide profiles among secondary students through a gender perspective. Their cluster analysis considered sociodemographic, academic and technological indicators alongside academic aspirations and digital inclusion. The authors found that the relationship between social background, technology and academic aspirations appeared in the profiles of boys but not in the same way among girls. This suggests that digital inequality should not be treated as one uniform experience: students' backgrounds and gender may intersect with technology use and STEM-related aspirations.

A related study by Niño-Cortés, Grimalt-Álvaro, Lores-Gómez and Usart (2023) surveyed 3,249 students in the second and fourth years of ESO and the second year of Bachillerato. It examined self-perceived digital competence and attitudes towards technology. The authors reported differences between boys and girls, with boys reporting higher digital competence and more positive attitudes. Perceived competence increased with school year, while attitudes among girls declined significantly. Because the study measures self-perceptions and attitudes, it is not a direct measure of STEM achievement. It nevertheless shows that the digital divide includes confidence and attitudes, not just equipment.

Together, these studies point in the same general direction but answer different questions. The PISA-based study of ICT use finds a subject-specific relationship with outcomes; the computational-thinking study highlights socioeconomic context and home ICT use; and the student-profile studies show that digital competence, attitudes and aspirations may differ across groups. The strongest conclusion is not that technology always improves learning, but that access and use should be examined alongside the social and educational conditions in which they occur.

3. Data and Methods

The review builds on the search strategy developed in the previous Atlas assignments. The focus is secondary education in Spain, with attention to technology access, digital competence, educational technology use and STEM-related outcomes. Searches included Google Scholar and ERIC, alongside Spanish-language sources found through Dialnet and journal platforms. Institutional sources included OECD and PISA publications, the Spanish National Statistics Institute (INE), the Spanish National Institute for Educational Evaluation (INEE), UNESCO and the European Commission.

The evidence is organised into three groups: official descriptive evidence, including household ICT statistics and ICILS 2023; empirical research using Spanish student or school data; and policy material that helps explain how digital education is organised in Spain. These sources are not interchangeable. Household access, digital competence, ICT use at school and academic achievement are different measures.

This is a focused literature review, not a systematic review or a meta-analysis. It does not combine results into a single effect size. The studies use different samples, years, measures and research designs, so their findings are compared cautiously. Associations in survey or assessment data are not treated as proof that access to technology independently causes higher achievement.

4. Results

4.1 Household access and student access

Official figures show that household connectivity is widespread in Spain, but access to a computer is not universal. The Spanish National Statistics Institute (INE) reported that in 2025, 97.4% of households with at least one member aged 16–74 had broadband internet access and 83.8% had some type of computer, including a desktop computer, laptop or tablet (INE, 2025). These are household-level indicators for the population covered by the survey, not direct measures of whether every secondary-school student has a suitable device available when needed. They establish the wider context, but should not be interpreted as proof that students have equal opportunities to use technology for schoolwork.

ICILS 2023 provides more directly relevant evidence because it assesses students' computer and information literacy and asks about resources available at home. In Spain, 63% of students reported having a good-quality internet connection at home, 68% had at least two computers at home, and 57% reported access to a computer for schoolwork. The corresponding EU figures were 62%, 74% and 64%, respectively (INEE, 2024). These indicators show why internet connectivity alone is an incomplete measure of digital inclusion: students may have broadband but lack a computer they can use independently for school assignments.

A student may be able to connect to the internet on a phone but still find it difficult to complete a long assignment, work with a spreadsheet or use specialised software. Another student may share a computer with family members. A household-level statistic cannot capture every situation, but the ICILS indicators suggest that the quality and type of access remain relevant.

TABLE I. HOUSEHOLD ICT ACCESS AND SELECTED ICILS STUDENT-ACCESS INDICATORS.

Indicator	Spain	EU average	Source / note
Households with broadband internet (2025)	97.4%	—	INE; household-level indicator
Households with a computer (2025)	83.8%	—	INE; household-level indicator
Students reporting good-quality home internet (ICILS 2023)	63%	62%	ICILS 2023 Spanish report
Students with a computer for schoolwork (ICILS 2023)	57%	64%	ICILS 2023 Spanish report
Students with at least two computers at home (ICILS 2023)	68%	74%	ICILS 2023 Spanish report

Sources: INE (2025) and INEE (2024). Household and student indicators describe different populations and units of analysis and should not be compared as though they measured the same thing.

4.2 Digital competence and socioeconomic differences

ICILS 2023 presents a more balanced picture of digital competence in Spain. The average Spanish score was 495 points, above the ICILS participant average of 476 and the EU average of 493. Yet 44.4% of students did not reach Level 2 in computer and information literacy. The socioeconomic performance gap was 69 points in Spain, compared with 79 points for the EU average reported by the Spanish National Institute for Educational Evaluation (INEE, 2024). These findings are not contradictory: Spain performs slightly above the EU average overall, while a substantial minority of students still has not reached Level 2 and outcomes remain associated with socioeconomic background. This contrast—strong average performance alongside internal inequality—is central to the paper's argument.

ICILS also reports differences associated with the quality of home access. In Spain, the difference in digital-competence performance between students with a good-quality internet connection and those without internet at home or with a slow or interrupted connection was 22 points. Students with at least two computers at home scored 32 points higher than students with fewer computers, while access to a computer for schoolwork was associated with a 34-point difference. These are descriptive associations, not estimates of the effect of providing a device or connection. Read alongside the national average and socioeconomic gap, they suggest that both the quality of access and the circumstances in which students use technology matter.

4.3 ICT use, STEM-related outcomes and gender

The empirical studies show that the relationship between ICT use and learning differs by outcome. Fernández-Gutiérrez, Giménez and Calero (2020), using PISA data from 2009, 2012 and 2015 across 13 Spanish autonomous communities, found a positive association between school ICT use and science scores, while the estimated effects for mathematics and reading were not statistically significant. Ortega-Rodríguez (2025), analysing PISA 2022 data from 30,800 Spanish secondary students in 966 schools, identified socioeconomic context and home ICT use among the predictors of computational-thinking performance. These findings support a subject-specific interpretation rather than the claim that more technology necessarily improves every kind of academic result.

Gender should also remain part of the analysis. Ortega-Rodríguez (2025) reports a gender difference in computational-thinking performance, while Niño-Cortés et al. (2023) examine gender differences in self-perceived digital competence and attitudes towards technology. In that survey of 3,249 students, boys reported higher self-perceived competence and more positive attitudes, with differences varying by school year. These measures are not interchangeable: performance on a computational-thinking assessment, confidence in one's digital skills and attitudes towards technology capture different aspects of the digital divide. The studies therefore suggest that gender is relevant, but do not support treating it as one single gap across all digital and STEM outcomes.

4.4 Regional differences within Spain

The Spanish ICILS report identifies regional differences in socioeconomic gaps in digital competence. The largest gaps reported for the regional samples were in Catalonia (77 points) and Cantabria (76 points). The smallest gap was in Asturias (57 points), followed by Castilla y León and the Canary Islands (58 points each), compared with 69 points for Spain overall (INEE, 2024). These figures refer to differences in average digital-competence performance by socioeconomic background. They should not be confused

with the separate indicator showing the percentage of students who did not reach Level 2. The results reinforce the paper's main point: a national average can coexist with meaningful inequalities within Spain.

5. Discussion

These results help explain why access and competence need to be measured separately. Students need more than a connection to find and evaluate reliable information, use software and complete digital tasks. At the same time, competence is not simply an individual responsibility: access to devices, opportunities to practise, teacher support and school resources can affect whether those skills develop.

Digital competence can support work with data, scientific information, simulations and computational tasks. However, ICILS measures computer and information literacy, not STEM participation itself. Its results help explain a possible pathway between access and STEM-related learning, but they cannot establish that the observed digital-competence gap produces a particular gap in mathematics, science or future STEM choices.

The regional structure of Spanish education matters because responsibility for digital infrastructure and many educational tools is distributed across autonomous communities. According to the OECD (2023), regional authorities provide much of the digital infrastructure used in schools, while the central government also provides resources and supports national initiatives. The report notes that differences in the tools and resources provided across regions create a challenge for equitable access and use.

This context matters because national averages can hide differences in how digital education is organised and experienced. Two regions may have similar household internet-access rates but different school equipment, learning platforms, teacher support or strategies for lending devices to students. These differences could affect opportunities to use technology for learning, even if household connectivity is broadly high.

However, decentralised governance is not itself evidence that regional inequality in STEM outcomes has already been demonstrated. To make that claim, a study would need comparable regional indicators for access, school use and student outcomes over the same period. The sources support the importance of regional policy context, but this review does not calculate a new regional ranking or claim that one community performs better because of its digital policy.

6. Recommendations

The findings suggest that digital-inclusion policies should look beyond the number of devices distributed. Reliable internet, access to a suitable computer for schoolwork, students' digital competence and the way technology is used in lessons should all be considered. The ICILS findings are useful because they show differences associated with connection quality and computer access, alongside a large socioeconomic gap in digital competence.

For schools, providing equipment is an important step for students who lack it, but it may not be enough on its own. Students also need opportunities to use digital tools for meaningful learning, and teachers need support to integrate those tools into lessons. The PISA-based evidence that ICT use was related to science outcomes but not significantly to mathematics or reading in one study suggests that the type of activity and the subject matter deserve attention.

Future research could combine student-level data on device access and connection quality with measures of digital competence, socioeconomic background, school resources and direct STEM outcomes. A longitudinal design would help researchers examine whether changes in access are followed by changes in competence or participation. Regional comparisons could also be informative, but only if indicators, years and student populations are comparable.

7. Limitations

The main limitation is that the studies do not measure exactly the same things. INE describes household ICT access, ICILS assesses digital competence and related home resources, and PISA-based studies analyse academic or computational-thinking outcomes. These sources complement each other, but their figures cannot be combined into one estimate of the effect of access on STEM learning.

The studies also rely mainly on observational data. Even when a statistical model accounts for several student and school characteristics, it cannot automatically rule out all alternative explanations. The results therefore support careful statements about associations and patterns, not a claim that access alone causes better outcomes.

STEM participation is broader than test scores. Mathematics, science and computational thinking are relevant outcomes, but they do not fully capture course choices, extracurricular participation or later study and career decisions. Some studies examine attitudes or self-perceived competence rather than assessed skill, so those findings need to be interpreted according to what was measured.

8. Conclusion

This review examined how access to technology relates to STEM learning and education among secondary-school students in Spain. INE's 2025 data show that 97.4% of households with at least one member aged 16–74 had broadband access and 83.8% had some type of computer. These are household-level figures and do not guarantee that every student has a suitable device available for schoolwork. ICILS 2023 provides a more student-focused picture: Spain scored 495 points in computer and information literacy, above the international participant average of 476 and the EU average of 493. However, 44.4% of students did not reach Level 2, and the socioeconomic performance gap was 69 points, below the EU gap of 79 points.

Empirical studies also show that the relationship between technology and learning is not the same for every outcome. Research using PISA data found a positive effect of increased school ICT use on science results, but no statistically significant effect on mathematics or reading. Other studies connect socioeconomic context and home ICT use with computational-thinking performance and show differences in digital competence and attitudes among student groups.

The answer to the research question is therefore qualified. Access to suitable devices and reliable connectivity can shape students' opportunities to participate in digital learning, including activities relevant to STEM. However, access alone does not explain differences in competence, achievement or aspirations. The purpose of technology use, students' skills, socioeconomic circumstances and regional school policies also matter. The evidence supports improving equitable access while evaluating how technology is used, rather than assuming that providing more technology will automatically improve STEM outcomes.

References

Fernández-Gutiérrez, M., Giménez, G., & Calero, J. (2020). Is the use of ICT in education leading to higher student outcomes? Analysis from the Spanish Autonomous Communities. Computers & Education, 157, 103969. https://doi.org/10.1016/j.compedu.2020.103969

Instituto Nacional de Estadística. (2025, November 20). Survey on Equipment and Use of Information and Communication Technologies (ICT) in Households. Year 2025. https://www.ine.es/dyngs/Prensa/en/TICH2025.htm

Instituto Nacional de Evaluación Educativa. (2024). ICILS 2023: Informe español. Ministerio de Educación, Formación Profesional y Deportes. https://www.educacionfpydeportes.gob.es/inee/evaluaciones-internacionales/estudio-icils-2023.html

Niño-Cortés, L. M., Grimalt-Álvaro, C., Lores-Gómez, B., & Usart, M. (2023). Brecha digital de género en secundaria: diferencias en competencia autopercibida y actitud hacia la tecnología. Educación XX1, 26(2), 299–322. https://doi.org/10.5944/educxx1.34587

Niño-Cortés, L. M., Grimalt-Álvaro, C., Serrano, V., & Sanabria, I. Z. (2024). Brecha digital en la educación secundaria: perfiles del alumnado desde una perspectiva de género en disciplinas STEM. Aloma, 42(2), 59–69. https://dialnet.unirioja.es/servlet/articulo?codigo=10275478

OECD. (2023). Country Digital Education Ecosystems and Governance: A Companion to Digital Education Outlook 2023. OECD Publishing. https://doi.org/10.1787/906134d4-en

Ortega-Rodríguez, P. J. (2025). PISA 2022. Predictores del rendimiento en pensamiento computacional en Educación Secundaria en España. Revista de Educación a Distancia (RED). https://doi.org/10.6018/red.600641

Contributions

Nada Zagmouti selected the official statistics and peer-reviewed studies, extracted and compared the relevant INE and ICILS figures, and wrote the analysis. The author organised the evidence around household access, student access, digital competence and STEM-related outcomes, interpreted national and regional findings, and developed the conclusion that equitable access should be considered alongside digital skills, socioeconomic context and the educational use of technology.

Appendix A. Data Sources

INE household ICT statistics (2025): https://www.ine.es/dyngs/Prensa/en/TICH2025.htm

ICILS 2023 Spanish report, Chapter 2.5 (sociodemographic and regional differences): https://www.educacionfpydeportes.gob.es/inee/evaluaciones-internacionales/estudio-icils-2023/capitulo-02/distribucion-factores-sociodemograficos.html

OECD report on digital education ecosystems and governance (2023): https://doi.org/10.1787/906134d4-en

Fernández-Gutiérrez et al. (2020): https://doi.org/10.1016/j.compedu.2020.103969

Ortega-Rodríguez (2025): https://doi.org/10.6018/red.600641

Niño-Cortés et al. (2023): https://doi.org/10.5944/educxx1.34587

Niño-Cortés et al. (2024): https://dialnet.unirioja.es/servlet/articulo?codigo=10275478

Appendix B. Evidence Rules

1. Access is not the same as use: having a device or internet connection does not show how often or for what purpose a student uses it.
2. Use is not the same as competence: frequent technology use does not automatically mean that a student can use digital tools effectively.
3. Competence is not the same as outcomes: digital-literacy scores and STEM achievement measure different things and should not be treated as interchangeable.
4. Association is not causation: differences between students with different access levels may also reflect socioeconomic background, school resources or other factors.
5. Keep the unit of analysis clear: household-level INE statistics should not be described as if they directly measure each student's access.
6. Trace figures to the original source: when a report or chart is cited, check the underlying report, table or questionnaire where possible, and state the year and population measured.